Spatio-Temporal correlated Network State Prediction and Dynamic Routing for Satellite Networks

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***Abstract*—Most existing routing algorithms for Low-Earth-Orbit (LEO) satellite networks neglect the spatio-temporal features inherent in the network states, leading to suboptimal performance in scenarios with dynamic network topologies. In this paper, we propose a Spatio-Temporal Graph Attention Network (STGAN) architecture for the extraction of spatio-temporal features from satellite network states. Building upon this, we propose a State Prediction based Dynamic Routing Algorithm (SP-DRA), which combines STGAN with the enhanced Shortest Path First algorithm (eSPF). SP-DRA utilizes STGAN to predict future network states by exploiting the spatio-temporal correlations of historic observations. Based on state predictions, each link is assigned with a delay-based prediction weight, which is used as the input for eSPF. The proposed eSPF dynamically selects the path with the smallest prediction weight to facilitate congestion avoidance and reduce transmission delay. Simulation results show that our STGAN architecture provides up to 18.6% prediction accuracy improvement than the existing deep learning-based methods, namely STGCN and ST-MGAT. Meanwhile, the SP-DRA algorithm outperforms existing routing strategies including SPF, Explicit load balancing algorithm (ELB), and Satellite networks Link State Routing algorithm (SLSR) in terms of packet loss rate and average end-to-end delay. Specifically, the performance gains increase with network traffic.**

***Index Terms*—LEO satellite network, routing algorithm, state prediction, spatial-temporal analysis**

I. Introduction

As a complement to the ground network, the Low-Earth-Orbit (LEO) satellite networks will play an important role in the future 6G systems [1]. Several LEO satellite constellations have already been put into operation, including Starlink, OneWeb, and Kuiper [2]–[4]. The Quality of Service (QoS) in LEO satellite networks is highly dependent on the routing algorithm, which ensures efficient and reliable operation of the satellite communication network. However, the high dynamism of nodes, links, and topology in LEO satellite networks imposes unprecedented challenges for routing [5].

To address the mobility of satellites, traditional routing methods for satellite networks primarily adopt topology-control-based strategies, which can be categorized into virtual node-based, virtual topology-based, and coverage domain division-based approaches. Mauger et al. [6] proposed a method of virtual nodes (VN), which uses virtual nodes to map and manage the dynamic position changes of satellites, thereby simplifying routing complexity and enhancing network flexibility and stability. Werner et al. [7] proposed a discrete state routing algorithm, which uses inter-satellite link topology information to compute multiple alternative routing paths between any two satellites, following the principle of minimum path delay, within each time segment. Uzunalioğlu et al. [8] proposed a footprint handover rerouting protocol (FHRP), which leverages the unique coverage area characteristics of satellites to simplify the routing decision-making process. The existing virtual node-based, virtual topology-based, and coverage domain division-based routing algorithms adopted static routing strategies, which used the periodicity of satellite network to calculate the topology information and routing information in advance, simplifying the routing calculation to a certain extent. However, they failed to make routing decisions dynamically with the real-time state of satellite network, making it unable to deal with burst traffic and link congestion.

With the development of machine learning, it has been applied to the field of network routing, enhancing the performance of routing algorithms. Na et al. [9] proposed an Extreme Learning Machine-based Distributed Routing strategy (ELMDR), which considers the density of traffic distribution on the Earth's surface and uses extreme learning machine for traffic prediction. Zhao et al. [10] proposed a load-balancing routing algorithm based on Multi-Agent Deep Reinforcement Learning (MADRL) for LEO satellite networks. Each LEO satellite is considered as an agent that determines its forwarding direction based on the number of data packets in the queue of neighboring satellites and the congestion level of the links. Zuo et al. [11] proposed a Deep Reinforcement Learning based Intelligent Routing algorithm (DRL-IR), where each LEO satellite autonomously

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makes routing decisions to minimize end-to-end latency of data flows. However, most of the existing machine learning-based algorithms made routing decisions based on current and past states, ignoring the spatio-temporal correlation of node states in the operation cycle. The lag of the algorithms potentially leads to network congestion. The topology of satellite network changes periodically, and the states of nodes are correlated with the past states and the states of neighboring nodes. By analyzing the historical and current satellite states, it is possible to build an appropriate model to speculate the future satellite network states on the time and space scales.

In view of the above problems, we propose a State Prediction based Dynamic Routing Algorithm (SP-DRA) for LEO satellite networks based on network state prediction, aiming to achieve preemptive scheduling of data transmissions to avoid link congestion, reduce transmission delay and improve transmission reliability. The major contributions of this paper are as follows:

- We propose a Spatio-Temporal Graph Attention Network (STGAN) architecture for predicting LEO satellite network states. By analyzing the spatial-temporal correlation of node features, STGAN achieves accurate state prediction of satellite networks.
- We propose a dynamic routing algorithm SP-DRA, based on the proposed STGAN and an enhanced Shortest Path First (eSPF) for route decision in LEO satellite network. By dynamically adjusting routing decisions with the future state-based link weights, SP-DRA provides better performance than the traditional routing algorithms.
- Simulation results based on the Iridium constellation show that our proposed STGAN provides up to 18.6% prediction accuracy improvement, and our proposed algorithms can preemptively schedule network traffic to reduce end-to-end delay and decrease packet loss rate.

The rest of the paper are organized as follows. In Sec. II, the system model and the routing problem are described. In Sec. III, an introduction of the proposed SP-DRA is given in details. In Sec. IV, experiments and performance evaluation are presented. In Sec. V, the paper is concluded .

II. System model and problem formulation

A. System model

We consider a LEO satellite network as shown in Fig 1. The satellite network is modeled as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$, where $\mathcal{V}$ is the set of satellite nodes, $\mathcal{E}$ is the set of inter-satellite links (ISLs), $A \in R^{N\times N}$ represents the adjacency matrix, N is the num of satellite nodes. Each satellite has two inter-orbit links and two intra-orbit links. There are two situations that the inter-orbit links will not be considered :

1) Polar regions: In the polar region with large latitudes, it is difficult to establish a relatively stable inter-satellite link because of the large relative velocity of satellites in adjacent orbits.

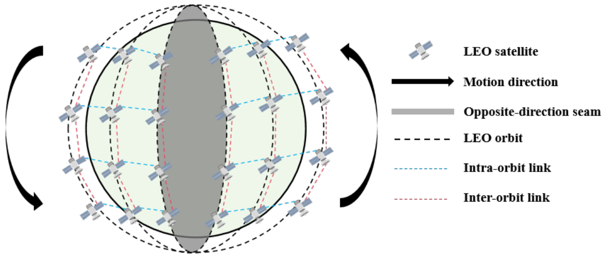

Fig. 1. Satellite Network Communication Model.

2) Orbit gap: The different direction of satellite motion leads to the existence of the orbit gap, where also no inter-orbit link between satellites exists.

B. Problem formulation

We decompose the state-prediction-based routing problem in satellite networks into two parts: the state prediction and the routing calculation.

1) State Prediction: As the satellites move and change their positions in orbit over time, their relative positions with other satellites and ground stations also change, making the satellite network status including link connection and data traffic load related to time. The core part of the state prediction is the function $\mathcal{F}$ as shown by

$$\hat{Y}_{\tau+1}, \cdots, \hat{Y}_{\tau+P} = \mathcal{F}\left(X_{\tau-T+1}, \cdots, X_{\tau}\right), \qquad (1)$$

which establishes a mapping relationship between the future states of the satellite network $\hat{Y}_P = (\hat{Y}_{\tau+1}, \cdots, \hat{Y}_{\tau+P})$ and the historical states $X_T = (X_{\tau-T+1}, \cdots, X_{\tau})$ at time τ.

2) Routing Calculation: The goal of the routing decision is to build the routing table based on predicted states and minimize the loss rate and end-to-end delay. The communication efficiency of LEO satellite network is mainly affected by three types of delays: propagation delay, transmission delay, and queuing delay. Considering the scenario of high-load LEO satellite networks, the propagation delay can be calculated in advance due to the deterministic orbit, and the transmission delay is relatively stable. At this point, node queuing delay becomes the critical factor affecting network performance. In this paper, we only consider the node queuing delay as the node features to predict. In environments with limited resources, accurately calculating queuing delay helps in evaluating and optimizing network performance.

In our discussion, each satellite maintains a First In First Out (FIFO) queue, which simulates the order of data packet processing in the reality. In the FIFO queue model, the queuing delay of a data packet is directly associated with its position in the queue. This principle reflects the shared nature of network resources, where each data packet must wait for the preceding packets to finish transmission before it can start its own transmission process. Assuming the packet f arrives the node u at time t, its queuing delay $T_u^Q(t)$ can be calculated by

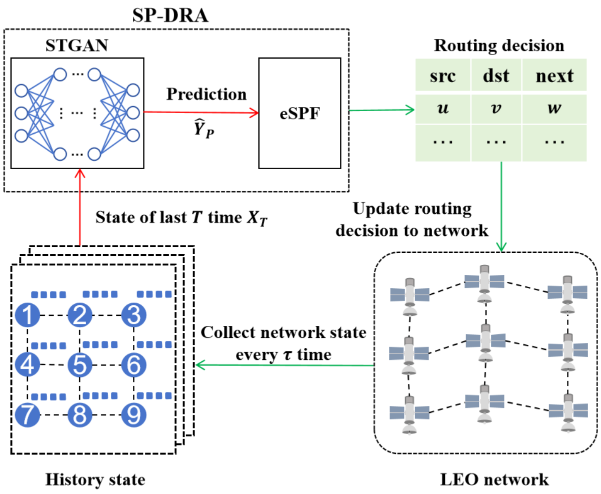

Fig. 2. Workflow of SP-DRA.

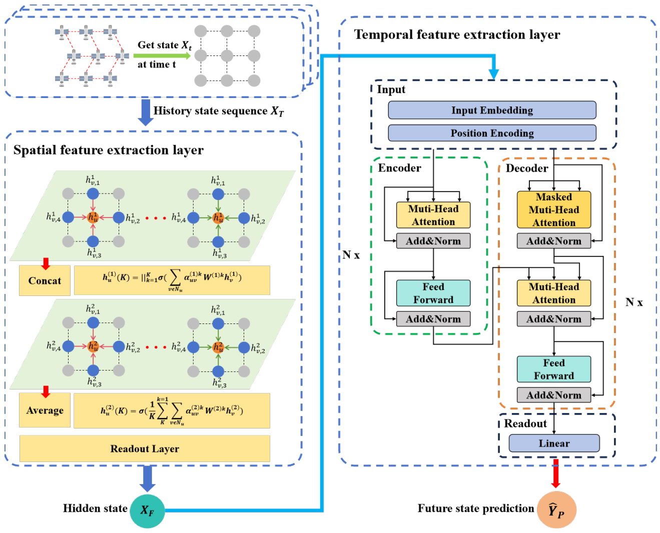

Fig. 3. Architecture of STGAN.

$$T_u^Q(t) = p(t) \cdot \frac{P_{ave}}{C}, \tag{2}$$

where $p(t)$ is the packets-queuing length, P_{ave} is the packets size, and C is the capacity of link.

The propagation delay $T_{u,v}^{prop}$ between nodes u and v at time t can be calculated by

$$T_{u,v}^{prop}(t) = \frac{l_{u,v}(t)}{c}, \tag{3}$$

where $l_{u,v}(t)$ is the length of inter-satellite link (u, v) at time t, and c is the speed of light.

Then the link cost $L_{cost}(t)$ can be defined by

$$L_{cost}(t) = T_{u,v}^{prop}(t) + \alpha \cdot T_u^Q(t), \tag{4}$$

where α is an adjustment factor used to control the degree to which node congestion $T_u^Q(t)$ affects link costs.

For each moment t, the model representation can be obtained as $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, A_t, X_t)$, where $X_t = (x_t^1, \cdots, x_t^N) \in R^{N\times F}$ is the state matrix of nodes in LEO satellite network at time t, F is the features of each node.

III. State Prediction based Dynamic Routing Algorithm

In this section, we will introduce the SP-DRA, consisting of a satellite network state prediction module based on STGAN and a routing decision module based on eSPF algorithm.

A. Framework

The workflow of SP-DRA is shown in Fig 2. The state of LEO satellite network is collected every τ time slot. At each route deciding time slot, STGAN first takes the historical node states $X_T \in R^{N\times T\times F}$ of the past T time slots as input, and outputs a state prediction $\hat{Y}_P \in R^{N\times P\times F}$ for the next P time slots. Since the satellite positions can be pre-computed using Two-Line Element (TLE) data, the propagation matrix T^{prop} for the next P time slots can be calculated in advance. We can obtain the link weight matrix based on T^{prop} and $\hat{Y}_P$. After that, the routing decision module will calculate the routing table with the link weight matrix and dynamically update the routing decision to the network.

B. State Prediction

STGAN utilizes satellite network topology and historical node state features to learn its spatio-temporal features and predict the future states. STGAN consists of two core components: spatial feature attention layer and temporal feature attention layer. The architecture of STGAN is shown in Fig 3.

1) Spatial Feature Attention Layer: The spatial feature attention layer is constituted by Graph Attention network (GAT) layers. GAT introduces an attention mechanism that focuses on one-hop neighbor nodes. The attention mechanisms in GAT are used to dynamically adjust the fusion weights of neighbor nodes features, which are determined only by the attributes of the nodes themselves rather than the inherent structure of the graph. This design breaks through the limitations faced by traditional Graph Neural Networks (GNNs) such as Graph Convolutional Network (GCN) and simplifies the process of learning the weights of different neighbors. We introduce GAT to learn the spatial features of the dynamic graph structure of satellite networks.

Firstly, GAT calculate the importance coefficient between each vertex and its neighbors, as given by

$$e_{uv}^{(l)} = \vec{a}^T \left[W^{(l)} h_u^{(l)} \parallel W^{(l)} h_v^{(l)} \right], v \in \mathcal{N}_u, \tag{5}$$

where $e_{uv}^{(l)}$ represents the importance of neighboring nodes to the vertex, W is the parameter of linear transformation, and

$\mathcal{N}_u$ is the set of adjacent nodes of node u. $\vec{a}^T$ is the parameter of a mapping function $a(\cdot)$, which is used to map the high-dimensional features to a real number. The transformation process is implemented using a single-layer Feed Forward Network (FFN).

Secondly, to compare the relative importance of each neighbor node, GAT uses softmax function for correlation coefficient normalization, as given by

$$\alpha_{uv}^{(l)} = \frac{\exp\left(\text{LeakyReLU}\left(e_{uv}^{(l)}\right)\right)}{\sum_{i \in \mathcal{N}_u} \exp\left(\text{LeakyReLU}\left(e_{ui}^{(l)}\right)\right)}. \tag{6}$$

Based on the calculated attention coefficient, the features of the nodes are weighted and summed up, as given by

$$h_u^{(l)} = \sigma\left(\sum_{v \in \mathcal{N}_u} \alpha_{uv}^{(l)} W^{(l)} h_v^{(l)}\right). \tag{7}$$

For the first layer of graph attention layer, the input is $h^0 = X^g$, where $X^g \in R^{N \times (T*F)}$ is obtained by rearranging the three-dimensional feature X_T. For other layers, the input hidden state is $h^{l-1} \in R^{N \times F_{in}}$. The output hidden state is $h^l \in R^{N \times F_{out}}$, where F_{in} is the input features, and F_{out} is the output features of each node. After L layers, the hidden state is h^L, and a readout layer is used to rearrange the two-dimensional hidden state into three-dimensional features $X_h \in R^{N \times T \times H}$ after all, where H represents the output feature dimension of each node.

Algorithm 1: Training of STGAN

Input: Training set O, number of epoch E, number of iteration I, learning rate γ, batch size B
Output: best model

```
Initialize STGAN parameters θ;
for epoch = 1, ··· , E do
    for iter = 1, ··· , I do
        Extracting batch samples X ∈ R^{B×N×T×F}, Y ∈ R^{B×N×P×F} ;
        Calculate predicted value Ŷ ;
        Calculate loss L(θ) between predicted labels Ŷ and actual labels Y;
        Update model parameters θ using backpropagation and optimizer;
    end
end
```

2) Temporal Feature Attention Layer: We adopt Transformer to capture the temporal correlation of the satellite network states. Transformer is known as the sequence transformation model, which improves training speed and model efficiency through parallel computing. At the same time, with designed position encoding strategies, it ensures that the model can maintain temporal sensitivity and accuracy while processing sequence data. There are two main parts in Transformer: encoder and decoder. This kind of architecture provides an ideal solution for predicting future states, especially in scenarios where multiple variable sequence predictions need to be processed simultaneously.

The temporal feature attention layer consists of several Transformer layers. The input of the temporal feature attention layer is the output X_h of the spatial feature attention layer. The output of the temporal feature attention layer is a prediction $\hat{Y}_P$ of future states of nodes.

The encoder consists of identical layers of multi-head attention and FFN, each followed by a residual connection and normalization layer. The input of encoder is firstly subjected to positional encoding before being fed into the multi-head attention layer. The multiple attention heads are used to capture the correlation information in the input sequence. The attention scores can be calculated by using the self-attention mechanism as given by

$$\text{Attention}\left(Q, K, V\right) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V, \tag{8}$$

where Q is the query matrix, K is the key matrix, V is the value matrix, and d_k is the dimension of the key vector.

The results calculated by multiple heads are then concatenated and subjected to a linear transformation. The calculation result of the attention head i is as given by

$$\text{head}_\text{i} = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \tag{9}$$

where W_i^Q,W_i^K,W_i^V are weight matrixs of linear transformation. The results are concatenated and linearly transformed as given by

$$\text{MultiHead}(Q, K, V) = \text{Concat}\left(\text{head}_1, \cdots, \text{head}_\text{h}\right) W^O, \tag{10}$$

where W^O is the weight matrix of the concatenated result subjected to linear transformation.

The FFN layer applies the fully connected neural network to each position in the sequence, as given by

$$\text{FFN}(x) = \max\left(0, x \cdot W_1 + b_1\right) \cdot W_2 + b_2, \tag{11}$$

where W_1, b_1, W_2, b_2 are parameters that can be learned during training.

The decoder is composed of layers of masked multi-head attention, multi head attention, and FFN. The decoder uses a masking multi-head self attention to avoid the problem of information leakage caused by the network obtaining future states during training. It introduces a mask matrix M when calculating attention weights to ensure that the decoder can only rely on current and previous outputs when generating sequences, thus meeting the requirements of the autoregressive model.

In this paper, we remove the last softmax layer compared to the original Transformer model to adapt to time series prediction. The fully connected layer is reserved for outputting the state prediction. The training process of STGAN is shown in Algorithm 1.

Algorithm 2: The proposed SP-DRA

Input: History states at time $\mathcal{G}_T = (\mathcal{V}, \mathcal{E}_T, A_T, X_T)$, propagation matrix for next P time $T^{prop}(P)$
Output: Routing table R

Initialize link quality matrix W_E, path matrix Pa, route table $Route$, weight coefficient α;
Get the predicted state with STGAN $\hat{Y}_P \leftarrow STGAN(X_T)$;
Calculate weight matrix of links using predicted state $W_E \leftarrow w\left(\hat{Y}_P, \alpha\right)$;
for $k = 1, \cdots, |\mathcal{V}|$ **do**
  for $i = 1, \cdots, |\mathcal{V}|$ **do**
    for $j = 1, \cdots, |\mathcal{V}|$ **do**
      if $W_E[i][j] > W_E[i][k] + W_E[k][j]$ **then**
        Update shortest path $W_E[i][j] \leftarrow W_E[i][k] + W_E[k][j]$;
        Update middle node of the shortest path $Pa\,[i]\,[j] \leftarrow Pa[i][k]$;
      end
    end
  end
end
$R \leftarrow r(P)$

C. Routing Algorithm

SPF algorithm is widely used for finding the shortest path in a weighted graph, gradually building a path tree starting from the source node. In the satellite network scenario, we mainly consider multi-source shortest path algorithms such as Floyd algorithm. Floyd algorithm utilize dynamic programming to find the shortest path among multiple source nodes.

As shown in Algorithm 2, the proposed SP-DRA algorithm utilizes STGAN to predict the future states, which are integrated into the calculation of weight matrix of links. The eSPF calculates the routing paths for nodes in the network based on the link weights instead of distance, where the shortest path is defined as the path with the minimum sum of link weights. The function $w(\cdot)$ uses the current state and predicted state of nodes to obtain the weight matrix W of the edges in the network. The function $r(\cdot)$ is used to calculate the shortest path and output routing table R with the calculated path matrix P.

IV. Experiments

We build a simulation system using Python. The constellation used in the simulation is Iridium constellation. The other detailed parameters are shown in Table I. As for the network traffic, we set up five pairs of source and destination nodes in the network, and the traffic is adjusted by changing the number of sending packets per second. The STGAN in SP-DRA is implemented based on Pytorch. We set $T = 12$, $P = 1$, and $F = 2$ in our simulation. The hidden features H of the output of GAT is set to 480, and the numbers of layers of GAT and Transformer are set to 2.

We evaluated the performance of SP-DRA under different values of α. We also compared the proposed SP-DRA with other three classic routing schemes, SPF algorithm, SLSR algorithm and ELB algorithm.

1) SPF algorithm: The SPF algorithm is derived from Dijkstra's algorithm, which directly utilizes the network distance matrix to calculate optimal routing paths.

2) SLSR algorithm: Satellite networks link state routing algorithm not only considers the transmission delay, but also takes the expected queuing delay as the link cost [12].

3) ELB algorithm: Explicit load balancing algorithm is a local flow equalization algorithm [13]. When making the flow balance decision, the satellite nodes only need to be aware of the flow state of the surrounding local areas, rather than the flow state of the whole network.

TABLE I
SIMULATION PARAMETERS

Parameter	**Value**
Number of satellites	66
Polar boundary latitude	80°
Link capacity	25 Mbit/s
Packet size	2 KB
Node packet buffer length	200 packets
Packet query routing delay	1 ms
Network traffic load	4 Mbit/s-20 Mbit/s
Simulation time	6000 s
State collection time slot	10 s

TABLE II
MODEL PERFORMANCE

Models	MAE	$RMSE$	$MAPE$(%)
STGCN	1.61	5.17	6.31
ST-MGAT	1.48	4.76	5.59
STGAN	**1.01**	**4.21**	**3.65**

Before starting the testing, the STGAN model proposed in this paper underwent 300 rounds of training. We compared STGAN with other classic models, STGCN [14] and ST-MGAT [15] on the same data set obtained through the simulation. Fig 4 shows that the training convergence process of our model is shorter and the loss values are smaller in the valid set.

To compare the performance of the proposed algorithm with the aforementioned baseline algorithms across various metrics, the trained model is tested, and other conditions such as packet requests and satellite network initialization are consistent. As shown in Table II, the proposed STGAN achieves 11.5% improvement compared with ST-MGAT and 18.6% improvement compared with STGCN on our simulation dataset.

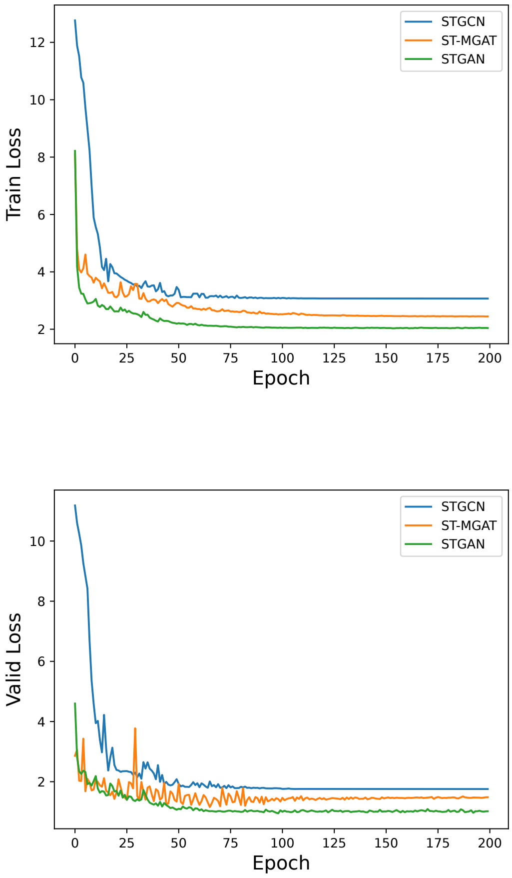

Fig. 4. Comparison of train loss and valid loss.

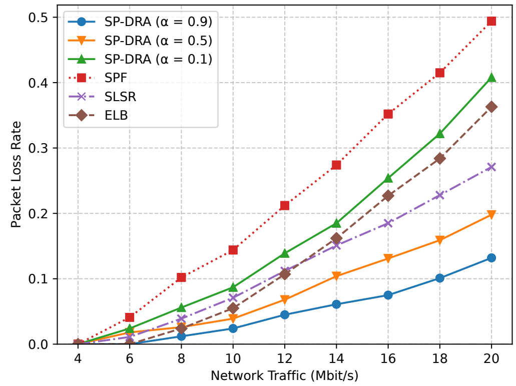

Fig. 5. Packet loss rate with different satellite network traffic load.

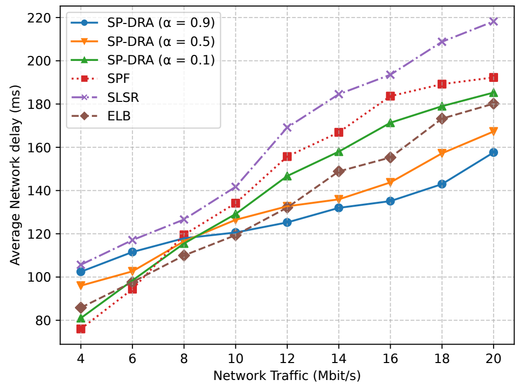

Fig. 6. Average end-to-end delay with different satellite network traffic load.

Fig 5 shows the trend of packet loss rate of different algorithms as the satellite network load changes. SPF has the highest packet loss rate because it only selects the shortest path for transmission, resulting in a highly concentrated traffic distribution. SLSR selects the path with the lowest total cost by considering information such as queuing delay, geographical location, and distributes traffic to the entire network, which greatly reduces the packet loss rate compared to SPF. However, repeatedly selecting the link with the lowest total cost for data transmission can cause certain routing oscillation problems. ELB performs local traffic offloading and alleviates packet loss during low load conditions. The ability to distribute is insufficient under high load conditions, leading to a noticeable reduction in performance. The SP-DRA ($\alpha = 0.9$) and SP-DRA ($\alpha = 0.5$) outperform other algorithms in terms of packet loss rate with the increase of network traffic. However, the SP-DRA ($\alpha = 0.1$) performs poorly in the simulation primarily due to the small attention coefficient for queueing delay, which causes the routing decisions to converge towards SPF.

Fig 6 shows the comparison of average end-to-end delay. At low traffic loads, the results of SPF and ELB are at low values than others because both algorithms consider end-to-end delay as the optimization goal and can achieve good performance when there is rare queuing delay in the network. SLSR shows a higher average delay than other algorithms because it requires packets to travel through more links in order to maintain link utilization. The SP-DRA ($\alpha = 0.1$) exhibits performance comparable to SPF. The reason is the queuing delay with low adjustment factor α has limited impact on routing decision. The performances of SP-DRA ($\alpha = 0.9$) and SLSR are close, which is because the data packets need to take a detour to avoid congestion when considering the queuing delay. When the load becomes heavier, the queuing delay increases, leading to the increase of average delay of SPF, ELB and SP-DRA ($\alpha = 0.1$). SP-DRA ($\alpha = 0.9$) and SP-DRA ($\alpha = 0.5$) achieve lower average latency compared to other algorithms because they make predictions rather than relying on statistical expectations, and update routing decisions at shorter time intervals, which brings the timeliness and forward-looking nature of their routing decisions.

V. Conclusion

In this paper, we have proposed a STGAN architecture and a SP-DRA for routing decision. GAT and Transformer are introduced to the proposed STGAN architecture to extract the spatio-temporal features of the satellite networks. The eSPF algorithm is used in SP-DRA, which uses the predicted states from STGAN for routing decision. Experiments show that STGAN outperforms existing prediction model STGCN and ST-MGAT, and the proposed SP-DRA has better performance compared with SPF, ELB and SLSR with the increase of network traffic. For future work, we intend to integrate the link states of the satellite network and other relevant data to improve the performance of the proposed routing algorithm.

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